

Presentation of Cleaned Dataset, Data Cleaning and Quality Report

Presented By- Impact Makers (Team 5)

Data Cleaning Log

Comprehensive step-by-step documentation of data transformation protocols applied to ensure integrity, uniformity, and analytical readiness.

- Column Standardization: Iterated through all variables to convert nomenclature to lowercase and replace spaces with underscores. Rationale: Ensures programmatic consistency and prevents syntax errors downstream.
- Placeholder Consolidation: Replaced ambiguous 'Unknown' string artifacts with standardized Null (NaN) values. Rationale: Prevents categorical skewing and accurately reflects true missingness for rigorous statistical evaluation.
- Datetime Casting & Imputation: Coerced temporal dimensions into structured datetime formats, deploying forward-fill strategies for continuity. Rationale: Enables precise time-series analysis and cohort tracking.
- Redundancy Elimination: Scanned full dimensionality for duplicated records and purged them. Rationale: Mitigates analytical bias, prevents overcounting, and optimizes storage efficiency.
- Textual Sanitization: Stripped leading and trailing whitespace across all stringent object-type variables. Rationale: Unifies structural formatting, resolving hidden inconsistencies critical for deterministic matching and BI integration.

Data Comparison Summary

The dataset underwent rigorous structural consolidation, yielding highly optimized metrics suitable for advanced indexing and BI modelling. The analytical summary delineates definitive quantitative improvements:

- Dimensionality Pipeline: Contracted from 7543 records to 7543 distinct records.
- Aggregate Missing Values: Modified from 371657 sparse entries to 366711 controlled null architectures.
- Duplicate Excision: Eliminated 0 redundant entries (Current Duplicate Count: 0).

Synopsis: Structural anomalies and duplicative inflation were successfully neutralized. Data coherence across categorical taxonomies and chronologies is strictly reinforced, optimizing data for direct deployment into dashboarding ecosystems.

Data Transformation Showcase (Sample)

RAW DATA SAMPLE (Pre-Cleaning - First 4 Columns):

Reference_ID	Given_Name	Last_Name	College
45405320	Deeksha	Reddy Bhumireddy	INDIA - Osmania University - Bachelor's Degree
858032003	Pearl Ashok	Kumar Patel	INDIA - Manipal University Jaipur - Bachelor's Degree

CLEANED DATA SAMPLE (Post-Cleaning - First 4 Columns):

reference_id	given_name	last_name	college
45405320	Deeksha	Reddy Bhumireddy	INDIA - Osmania University - Bachelor's Degree
858032003	Pearl	Ashok Kumar Patel	INDIA - Manipal University Jaipur - Bachelor's Degree

Output Deliverable of Cleaned Dataset

Dataset Certification & Finalization:

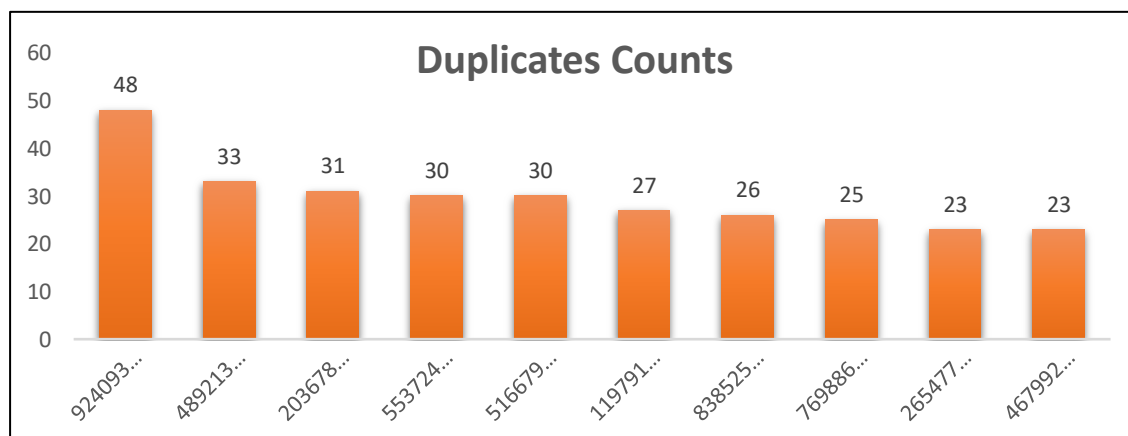
- Integrity Verification: Dataset is structurally sound, void of row-level duplication, and categorically reliable.
- Distribution: Successfully packaged and exported locally as 'cleaned_dataset_final.csv'.
- Infrastructure Ready: Standardized dimensions seamlessly adhere to best practices for Power BI, Tableau, and SQL integration schemas.

Here is a link of Cleaned Dataset: https://drive.google.com/file/d/1uiquQGzrbsg_-sto0q3s15OagCA905Fw/view?usp=drive_link

Visualization Before Cleaning

Top Duplicates Records Before Cleaning

Reference_ID	Counts
924093519	48
48921321	33
20367881	31
553724526	30
516679919	30
11979156	27
838525490	26
769886402	25
265477508	23
467992191	23

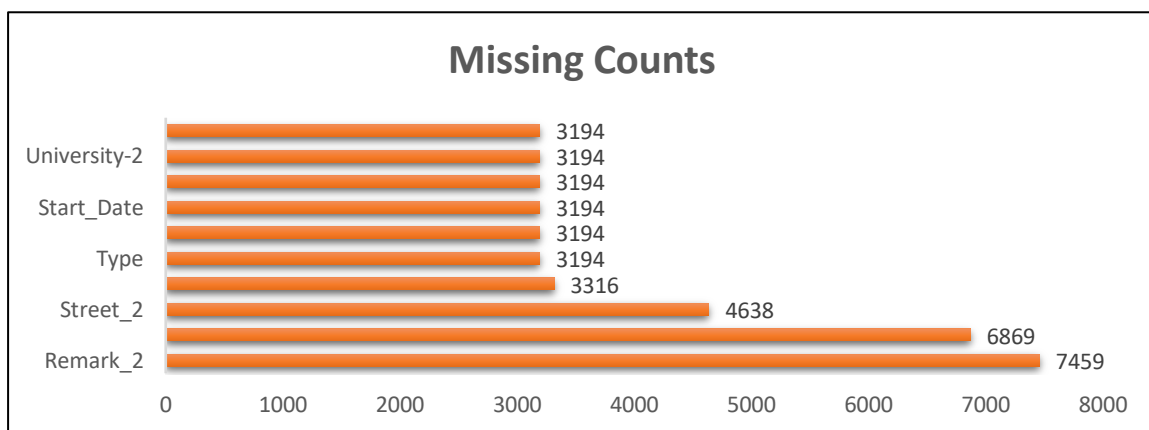


This chart highlights the Reference_ID values that appear most often in the dataset. This suggests that some records were entered multiple times.

Missing Values Before Cleaning

Column Name	Missing Count
Remark_2	7459
Remark	6869
Street_2	4638
Name	3316
Type	3194
University-3	3194
Start_Date	3194
Recieved_At-2	3194
University-2	3194
Date_of_Contact	3194

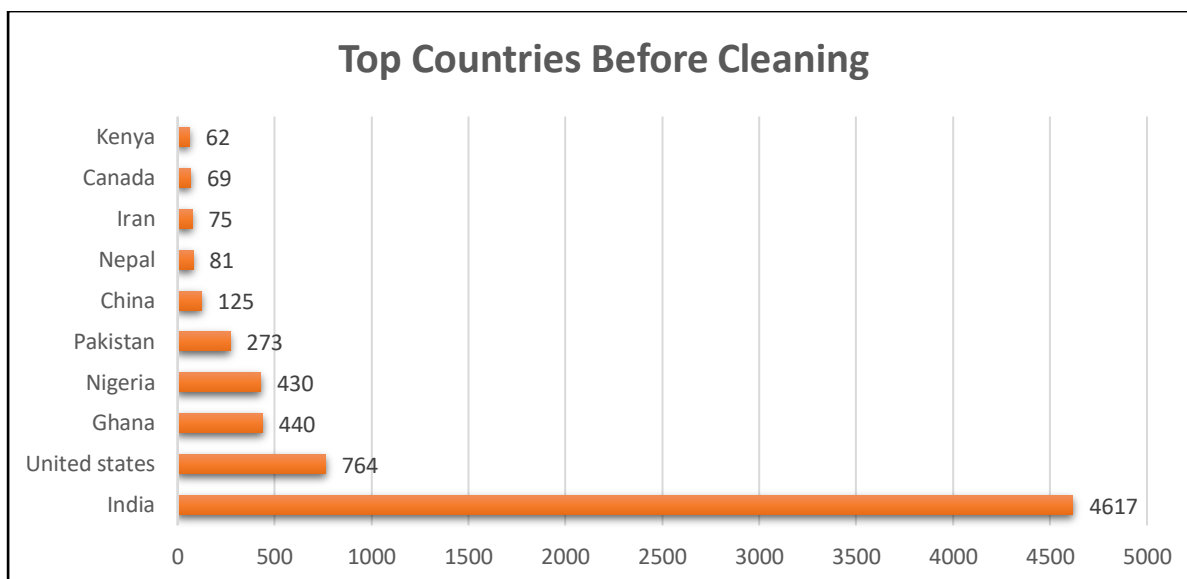
This table shows the columns with the highest number of missing values in the raw dataset. It makes it clear that some parts of the data are heavily incomplete.



This chart shows the number of missing values across selected columns in the dataset before cleaning. It indicates that some columns contain a significant amount of incomplete data.

Top Countries Before Cleaning

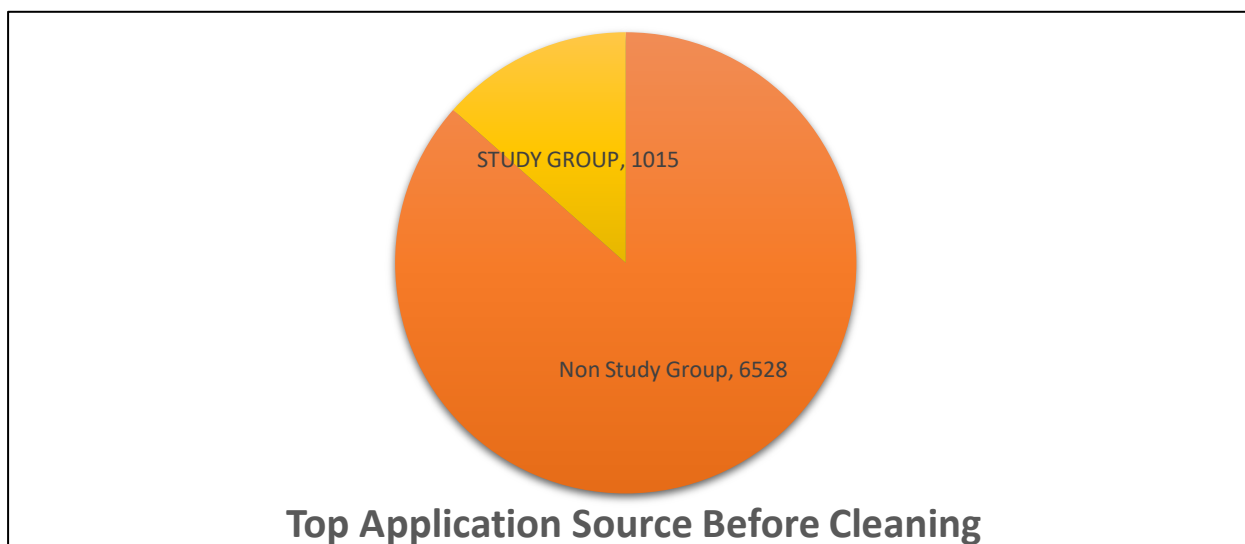
Country	Count
India	4617
United states	764
Ghana	440
Nigeria	430
Pakistan	273
China	125
Nepal	81
Iran	75
Canada	69
Kenya	62



This chart shows how data are distributed across different countries. It helps give a quick view of where most of the entries are coming from before any cleaning is carried out.

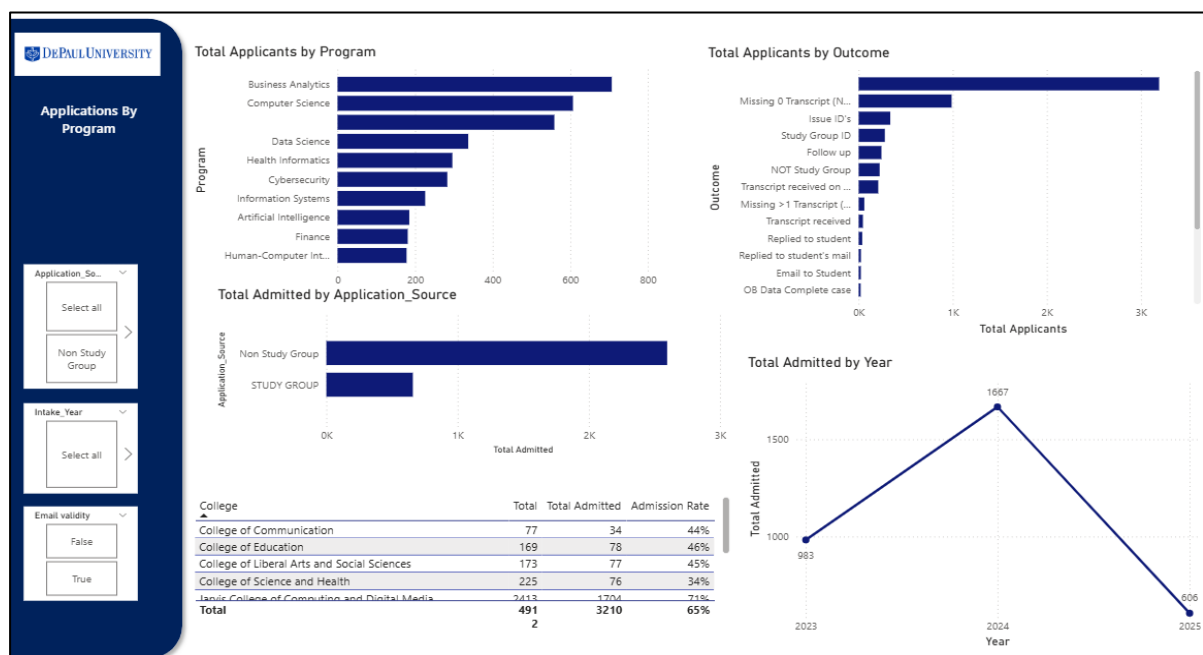
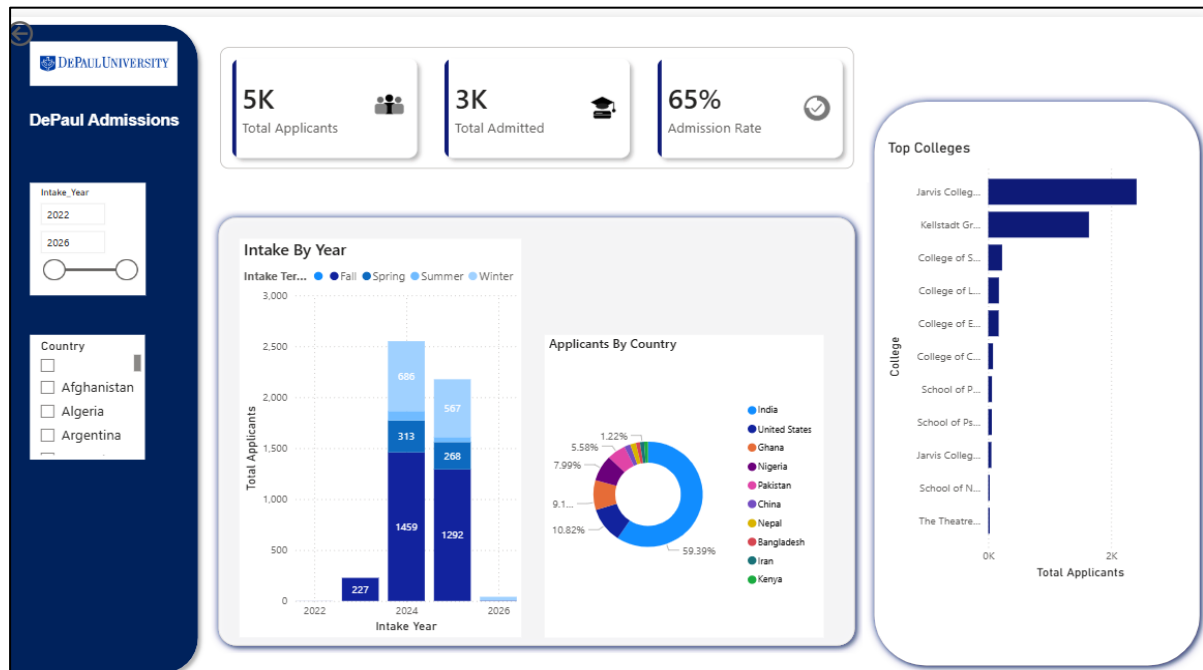
Application Source Before Cleaning

Application Source	Count
Non-Study Group	6528
STUDY GROUP	1015



This chart shows the source that contributes the largest number of entries.

Visualization After Cleaning



Key Findings

- 5,000 unique applicants with a 65% admission rate (3,210 admitted)
- India accounts for 59% of all applicants, making it the dominant source country
- 2024 Fall was the peak intake period across all years
- Business Analytics and Computer Science are the most applied-for programs
- Missing Transcripts is the biggest pipeline bottleneck, affecting 1,177 applicants
- Jarvis College of Computing and Digital Media recorded the highest admission rate at 71%

Short Summary Report

Data cleaning substantially improved the dataset's reliability and usability by removing inconsistencies and standardizing structure. Key steps included renaming columns for consistency, converting "Unknown" values to proper nulls, formatting datetime fields, and cleaning text errors. These changes ensure accurate representation of missing data, eliminate hidden inconsistencies, and make the dataset fully compatible with analytical tools.

As a result, the dataset is now more dependable for analysis and reporting. Cleaned data enables accurate insights, such as identifying true bottlenecks (e.g., missing transcripts), reliable country-wise distributions, and correct admission rates. It also supports valid time-based analysis, like detecting peak application periods.

Overall, cleaning transformed the dataset from error-prone to analysis-ready, improving both the quality of insights and confidence in decision-making.

Initial Insights (Impacts on Analysis Outcomes)

Data cleaning improves accuracy and trust in analysis by ensuring missing values, categories, and dates are handled correctly. This leads to more reliable metrics (e.g., admission rates), clearer identification of bottlenecks (like missing documents), and accurate trend analysis (such as peak intake periods). Overall cleaning reduces bias and errors, enabling more meaningful and actionable insights.

Note: -

Here is the link containing all relevant resources, including the cleaned dataset, data comparison summary, data profile, and the complete cleaning process. Please refer to it for detailed information and documentation.

https://docs.google.com/spreadsheets/d/1cH6kXGclqJZ6VpLtDVJXzY1iDij9r54-/edit?usp=drive_link&ouid=107074866250955083491&rtpof=true&sd=true